KNOWLEDGE SERIES · 02

Show Your Work

The Project Controls Competency Framework

PCI AI · PROJECT CONTROLS INSTITUTE GLOBAL
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AI proposes. The professional disposes.

Contents

1. Executive summary

Most competency frameworks fail for the same reason: they describe qualities rather than evidence. "Demonstrates strong analytical skills" cannot be assessed, cannot be developed deliberately, and cannot be defended in a promotion discussion. It is an adjective wearing a lanyard.

This framework takes a different position. **A competency is a claim about what a person can produce and defend.** If there is no artefact and no defence, there is no competency — whatever the training record says.

That standard is demanding, and it is the point. It makes competence assessable, developable and portable.

Six findings frame what follows.

One. Competence is evidenced, not asserted. Section 4 sets the standard: an artefact, a defence of the choices inside it, and independent verification. Three parts, all required.

Two. Certification reaches three registers of competence; the workplace pays for a fourth. An examination can test recall, application and judgement. It cannot test stewardship — the ability to set a standard, develop others and be accountable for a system. Any organisation that treats a credential as the whole competency picture will systematically under-develop its senior layer. Section 5 is explicit about this limit.

Three. Every proficiency level has a characteristic failure, and naming it is more useful than naming the strength. A Level 2 professional applies a correct method to a situation whose assumptions have broken. A Level 4 professional over-standardises. Section 6 gives all five.

Four. Team capability is set by coverage, not by average level. A team of eight strong cost engineers and one risk analyst is not a strong team; it is a strong cost function with a risk gap that will surface as unquantified contingency. Section 11 sets out the Coverage Rule and Section 12 makes it measurable.

Five. Four of the fourteen published competencies are AI competencies. Predictive analytics, AI-enabled project controls, responsible AI and human validation are 29% of the set. They are not an appendix to the discipline; they are part of it.

Six. AI creates a competency risk the profession has not yet priced. Judgement has always been a by-product of doing the work many times. AI removes much of that volume. If juniors never build the artefact by hand, the pathway that produced senior judgement breaks. Section 14 names this the Apprenticeship Gap and sets out four responses.

> **The one-sentence version.** Competence is not what a professional knows; it is what they can build, defend, and be checked on — and an organisation's capability is set by what it can cover, not by what it averages.

2. What a competency framework is for

A competency framework does four jobs. Most do only the first.

Job	What it looks like when done	What it looks like when it fails
Define the standard	A person can read the level above and know what to build	Adjectives that describe attitude rather than output
Assess an individual	Two assessors reach the same level independently	The level tracks tenure or likeability
Diagnose an organisation	The capability gap is named, located and sized	A heat map nobody acts on
Develop deliberately	A named artefact to produce next, with a verifier	A course catalogue

The three ways frameworks fail.

Failure one — adjectival drift. The framework describes personal qualities: collaborative, proactive, detail-oriented. These are real traits and they are not competencies, because two assessors will not agree on them and no one can practise them deliberately on a Tuesday.

Failure two — the tenure proxy. Levels get assigned by years served, because years are visible and competence is not. The framework becomes a description of the org chart it was meant to challenge.

Failure three — assessment without verification. Self-assessment and line-manager assessment both drift upward, in opposite directions and for different reasons. Without an independent check, a framework measures confidence and relationship, not capability.

> **The test of a competency framework.** Give the same person's evidence file to two assessors who have not spoken. If they return different levels, the framework is describing something other than competence.

> **Figure 1 — suggested diagram.** Four horizontal bars — Define, Assess, Diagnose, Develop — each split into a solid segment ("what most frameworks do") and a hollow segment ("what they leave undone"). The Define bar is nearly solid; Develop is nearly hollow. The visual asymmetry is the message.

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3. Why this matters

Two conditions make project controls competence unusually consequential, and unusually hard to see.

The first is delay. A weak controls decision does not fail immediately. An estimate built on the wrong class of accuracy, a schedule with no logic discipline, a contingency set by percentage rather than by quantified risk — none of these produce an error message. They produce a project that goes wrong eighteen months later, by which time the causal chain is long and the individual is elsewhere.

The second is plausibility. Publication 01 of this series set out the Seven Silent Failures: control failures that produce confident, internally consistent, entirely wrong reports. Every one of those failures is also a competency failure. Someone who does not know that a percentage complete supplied by the measured party is a commercial position rather than a measurement will produce a clean report and a wrong one.

Together these mean that in project controls, incompetence is quiet. It does not announce itself in the way a structural or a safety failure does. It shows up as a forecast that was always going to be wrong, produced by someone nobody had ever assessed.

The published evidence is consistent with this. Flyvbjerg's research programme found no measurable improvement in infrastructure cost-forecasting accuracy across seventy years of data — a period covering the introduction of critical path method, earned value management and enterprise systems (Flyvbjerg, Holm and Buhl, 2002; Flyvbjerg and Gardner, 2023). Tools improved. Outcomes did not. That pattern is more consistent with a capability constraint than a technology one.

> **Callout.** The industry has spent four decades improving what project controls professionals have to work with, and comparatively little on assessing what they can do with it. A framework that makes competence visible is the cheaper of the two interventions and has never been the funded one.

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4. The Evidence Standard

Original framework.

A competency claim at any level requires three things. All three, every time.

> **Artefact · Defence · Verification**
>
> **Artefact** — the thing produced. A time-phased budget. A quantitative schedule risk analysis. A variance bridge. A basis of estimate. Something that exists and can be read.
>
> **Defence** — the professional can explain why this method rather than another, which assumptions the output rests on, and what would change the answer.
>
> **Verification** — an independent assessor at Level 4 or above confirms both the artefact and the defence.

The rule this replaces. "Familiar with earned value management" is not a competency claim. "Has attended EVM training" is not a competency claim. "Five years in a cost role" is not a competency claim. Each is a proxy, and each is why competency frameworks lose credibility with the people they assess.

Why the defence matters more than the artefact. An artefact can be produced by copying a template, by inheriting a spreadsheet, or — increasingly — by asking a machine. None of those establish competence. The defence is what separates a person who produced the output from a person who understands it.

The defence is also where assessment gets efficient. A competent professional can defend their choices in ten minutes. An incompetent one cannot do it in an hour, because the reasoning was never there.

Three questions that constitute a defence. For any controls artefact:

1. **Why this method?** What alternatives existed and why was this one appropriate here?
2. **What is it resting on?** Name the assumptions, and say which are load-bearing.
3. **What would change it?** What new information would move the answer materially, and by roughly how much?

A professional who can answer all three owns the output. One who can answer only the first has learned a procedure.

> **Callout — the portability benefit.** Evidence-based competence travels. A professional who holds an evidence file of artefacts and defences can demonstrate capability to a new employer, a new sector or a new jurisdiction without relying on the reputation of their last one. This is the practical argument for the standard, and it belongs to the individual rather than the organisation.

> **Figure 2 — suggested diagram.** Three interlocking rings labelled Artefact, Defence, Verification, with the competency claim sitting only in the central intersection. Each pairwise overlap labelled with what it is not: Artefact + Defence without Verification = "unchecked claim"; Artefact + Verification without Defence = "someone else's work"; Defence + Verification without Artefact = "theory".

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5. The Four Registers of Competence

Original framework.

The examination blueprint for the credential writes items at three cognitive levels: recall, application and analysis. That is the right structure for an examination. It is not the whole structure of professional competence.

Register	The professional can	How it is evidenced	Can an examination reach it?
1 · Recall	State the definition, the principle, the standard	Written or oral response	**Yes**
2 · Application	Produce the artefact correctly by a defined method	The artefact	**Yes**
3 · Judgement	Choose between methods and defend the choice under challenge	The defence	**Partly** — through scenario and analysis items
4 · Stewardship	Set the method for others, verify their work, and carry accountability for the system	The record of decisions made, standards set and people developed	**No**

The honest limit. A certification body should say plainly what its examination can and cannot establish. An examination establishes registers 1 and 2 robustly and register 3 partially. It cannot establish register 4, because stewardship is demonstrated over time, across other people's work, and inside an organisation.

Why this matters commercially. Organisations that treat a credential as the complete competency picture will hire and promote against registers 1 to 3, and then be surprised that their senior controls layer cannot set a standard or develop anyone. The credential was never claiming to measure that.

The corollary. Register 4 is where the scarcity is. Registers 1 and 2 are increasingly served by tools; register 3 is where professional judgement lives; register 4 is where an organisation's capability is reproduced across time. A development programme that stops at register 3 produces excellent individual practitioners and no successors.

> **Callout.** The registers are cumulative in one direction only. Nobody reaches stewardship without judgement, and nobody develops judgement without having applied the method many times. Section 14 examines what happens when technology removes the application work.

6. The Proficiency Ladder

Original framework.

Five levels. Each is defined by four things — and the fourth is the one most frameworks omit.

Level	Name	Produces	Can be trusted to sign	Supervision	
L1	**Assisted**	Components of an artefact, to instruction	Nothing	Work checked before it leaves the desk	Follows the template past the point where it stops applying
L2	**Capable**	The complete artefact for routine cases, by a defined method	Routine outputs within a set method	Reviewed, not directed	Correct method, wrong situation — does not notice when the method's assumptions have broken
L3	**Independent**	The artefact for non-routine cases; adapts method with reasons	Own work on a project	Reviewed by exception	Defends own judgement well but does not surface the assumption to others; becomes a single point of dependency
L4	**Authoritative**	The method itself; arbitrates between approaches across projects	Others' work; the project's controls position	Accountable, not supervised	Over-standardises — imposes one method where genuine variation is warranted
L5	**Standard-setting**	The organisation's standard; develops L3s and L4s	The system, and its failures	Accountable to governance	Defends the standard against the evidence that it needs to change

Why the failure column exists. Every other column describes what someone can do, which helps with selection and promotion. The failure column describes what to watch for, which helps with supervision and with the professional's own development. It also removes the implication that higher is uniformly better — a Level 4 who over-standardises can do more damage than a Level 2 who is simply learning.

Reading the ladder correctly. Levels attach to a **competency**, not to a person. A capable professional will typically hold Level 4 in two or three competencies, Level 3 in several, and Level 1 or 2 in the rest. That profile is normal and healthy. A person claiming Level 4 across all fourteen competencies has almost certainly not been assessed.

The signing test. The fastest way to place someone is to ask what they can be trusted to sign. Nothing at all is L1. Their own routine work is L2. Their own judgement on a live project is L3. Someone else's work is L4. The standard everyone else signs against is L5.

> **Figure 3 — suggested diagram.** A five-rung ladder. Each rung carries its name and signing authority on the left. On the right, offset and in a contrasting tone, the characteristic failure — drawn as a small hook pulling backwards off the rung. The visual point: every level has a way of going wrong that belongs to it.

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7. The fourteen competencies

The Institute publishes fourteen competencies for the **PCI AI Project Controls Leader (PCL-AI)** credential. This framework assesses exactly those, grouped into five clusters for practical use.

Cluster	Competencies
A · Governance and measurement	1 Project controls governance · 6 Performance measurement
B · Cost and commercial	2 Cost management · 8 Commercial and contract controls
C · Time and forecast	3 Planning and scheduling · 4 Earned value management · 5 Forecasting and EAC
D · Risk	7 Project risk
E · Digital, analytics and AI	9 Predictive analytics · 10 AI-enabled project controls · 11 Digital reporting · 12 Automation · 13 Responsible AI · 14 Human validation

What each competency means in this framework

#	Competency	The professional can	Signature artefact
1	Project controls governance	Establish the control framework, cadence, thresholds and decision rights for a project	The controls plan and the decision-rights matrix
2	Cost management	Run the commitment-to-accrual-to-actual cycle and reconcile it to the finance ledger	The reconciled cost report with itemised differences
3	Planning and scheduling	Build and defend a logic-driven schedule, and control it through updates	The baseline schedule and its logic audit
4	Earned value management	Establish measurement rules, compute and interpret the indices honestly	The EVM measurement plan and a period analysis
5	Forecasting and EAC	Produce a forecast range with named assumptions and defend the selection	The EAC range with its basis
6	Performance measurement	Design the indicators, thresholds and exception rules that drive attention	The KPI set with defined thresholds
7	Project risk	Identify, quantify and link risk to contingency, and monitor drawdown	The QSRA and the contingency justification
8	Commercial and contract controls	Manage the commercial cycle, valuations, change and its cost effect	The change register with cost impact and the valuation
9	Predictive analytics	Build or commission predictive models and judge their fitness	The model, with validation evidence and stated limits
10	AI-enabled project controls	Apply AI to controls tasks at the correct level of autonomy	The workflow design with its assurance rung
11	Digital reporting	Build reporting that drives decisions rather than describing history	The exception-led report and its data lineage
12	Automation	Automate routine controls processes with controls that survive audit	The automated process with its control points
13	Responsible AI	Govern AI use — accountability, auditability, bias, disclosure	The AI governance standard and its audit trail
14	Human validation	Verify machine output competently and record the verification	The verification record with the reviewer named

Callout — competencies 13 and 14 are not soft. Responsible AI and human validation are the two competencies most often treated as policy rather than capability. They are capability. Verifying a predictive EAC requires knowing what a defensible EAC looks like, which is competency 5 at Level 3 or above. **You cannot validate what you could not have produced.**

8. The Competency Matrix

The operative content of this framework. Read across a row to see progression; read down a column to see what a level means across the discipline.

Cluster A — Governance and measurement

Competency	L1 Assisted	L2 Capable	L3 Independent	L4 Authoritative	L5 Standard-setting
1 Governance	Maintains the register and cadence set by others	Runs the agreed controls cycle on a project	Designs the controls plan for a project and sets thresholds	Sets controls governance across a portfolio; arbitrates disputes	Defines the organisation's controls standard and its assurance regime
6 Performance measurement	Populates defined KPIs	Produces the KPI set and flags breaches	Designs indicators and thresholds fit for the decision	Judges which measures drive behaviour and which distort it	Sets the measurement standard and retires measures that no longer earn their place

Cluster B — Cost and commercial

Competency	L1 Assisted	L2 Capable	L3 Independent	L4 Authoritative	L5 Standard-setting
2 Cost management	Codes and posts to instruction	Runs the cost cycle; prepares accruals from a defined rule	Owns a project's cost position and reconciles it to finance	Sets accrual and cut-off policy; resolves reconciliation disputes	Defines the cost control standard and the reconciliation regime
8 Commercial and contract controls	Maintains the change register	Prepares valuations and change entries to method	Owns commercial control on a project; assesses change cost impact	Advises on contract strategy and risk allocation across projects	Sets the commercial control standard and the change authority framework

Cluster C — Time and forecast

Competency	L1 Assisted	L2 Capable	L3 Independent	L4 Authoritative	L5 Standard-setting
3 Planning and scheduling	Updates activities and progress to instruction	Builds a schedule to a defined WBS and logic convention	Owns a project schedule; defends logic, float and critical path	Sets scheduling standards; audits others' schedules for logic integrity	Defines the planning standard and the schedule assurance regime
4 Earned value management	Collects progress data to a defined rule	Computes EV and the indices correctly	Sets measurement methods per work package and interprets indices	Judges where EVM applies and where it misleads; audits measurement integrity	Defines the EVM standard, including adaptive-delivery treatment
5 Forecasting and EAC	Prepares inputs to a forecast	Computes EAC by a specified formula	Produces a forecast range with named assumptions and defends selection	Arbitrates between competing forecasts; challenges optimism systematically	Sets the forecasting standard, including forecast scoring

Cluster D — Risk

Competency	L1 Assisted	L2 Capable	L3 Independent	L4 Authoritative	L5 Standard-setting
7 Project risk	Maintains the risk register	Runs qualitative assessment and expected-value calculations	Runs quantitative analysis and derives contingency from it	Sets risk appetite application; challenges contingency across a portfolio	Defines the risk standard and the link between risk and funding

Cluster E — Digital, analytics and AI

Competency	L1 Assisted	L2 Capable	L3 Independent	L4 Authoritative	L5 Standard-setting
9 Predictive analytics	Supplies and cleans data to instruction	Runs an existing model and reads its output	Builds or commissions a model and states its limits	Judges model fitness; rejects models trained on unfit data	Sets the analytics standard and the model-validation regime
10 AI-enabled project controls	Uses approved AI tools within a defined workflow	Applies AI to routine tasks and verifies output	Designs an AI-assisted workflow and assigns its assurance rung	Sets autonomy levels across the function by consequence	Defines the organisation's AI-in-controls standard
11 Digital reporting	Produces report components from a template	Builds the reporting pack to a defined design	Designs exception-led reporting fit for the decision and audience	Sets reporting standards; removes reporting that no one acts on	Defines the reporting standard and its data-lineage requirements
12 Automation	Operates automated processes	Configures automation within a defined pattern	Automates a controls process and builds its control points	Judges what should and should not be automated	Sets the automation standard and its audit requirements
13 Responsible AI	Follows the AI use policy	Applies disclosure and record-keeping rules correctly	Governs AI use on a project; maintains the audit trail	Sets AI governance across the function; handles incidents	Defines the responsible-AI standard and answers for it externally
14 Human validation	Checks output against a checklist	Verifies routine AI output competently	Verifies non-routine output and can reconstruct the correct answer independently	Sets verification requirements by task and consequence	Defines the validation standard and audits its application

> **Callout — how to use this matrix.** Do not score all fourteen at once. Assess the three or four competencies that a person's role actually requires at Level 3 or above, using the Evidence Standard. Scoring fourteen competencies for every professional produces a spreadsheet, not a diagnosis.

9. Mapping competence to the Body of Knowledge

Competencies describe what a professional does. The Body of Knowledge describes what they must know. The mapping shows where to study for a given capability gap.

#	Competency	Primary BoK domains	Reinforced in
1	Project controls governance	8 (lifecycle), 4 (reporting)	11, 12
2	Cost management	5 (cost control), 1 (accounting foundations)	2, 11
3	Planning and scheduling	10 (scheduling)	8, 9
4	Earned value management	6 (EVM)	3, 4, 9
5	Forecasting and EAC	6 (EAC family), 3 (forecasting)	12
6	Performance measurement	4 (performance and variance)	6, 10
7	Project risk	12 (risk)	3 (contingency), 10
8	Commercial and contract controls	7 (contracts, BoQ, invoicing)	2 (IFRS 15), 5
9	Predictive analytics	13.1, 13.2	3, 6
10	AI-enabled project controls	13.4, 13.5	Every domain's AI treatment
11	Digital reporting	4.3, 4.4	13.5
12	Automation	13.4, 11 (process cycles)	5
13	Responsible AI	13.6	13.7
14	Human validation	13.6	The competency being validated

Coverage check: fourteen of fourteen competencies map to Body of Knowledge content; none is orphaned.

The dependency worth noting. Competency 14 (human validation) has no fixed study route, because validating a machine's cost forecast requires competency 5, and validating its schedule analysis requires competency 3. Human validation is not a subject. It is the exercise of an existing competency against a new kind of input.

10. Role profiles

Required level by competency for common roles. **R** marks the competencies where the role carries primary accountability.

Competency	Controls Analyst	Cost Engineer	Planning Engineer	Commercial Controller	Controls Manager	Head of Controls
1 Governance	L1	L2	L2	L2	**L4 R**	**L5 R**
2 Cost management	L2	**L4 R**	L2	L3	L4	L4
3 Planning and scheduling	L2	L2	**L4 R**	L2	L3	L4
4 Earned value management	L2	L3	L3	L2	**L4 R**	L4
5 Forecasting and EAC	L2	L3	L3	L3	**L4 R**	L4
6 Performance measurement	L2	L3	L3	L2	**L4 R**	L4
7 Project risk	L1	L3	L3	L2	**L3 R**	L4
8 Commercial and contract controls	L1	L3	L1	**L4 R**	L3	L4
9 Predictive analytics	L1	L2	L2	L1	L3	**L3 R**
10 AI-enabled project controls	L2	L2	L2	L2	L3	**L4 R**
11 Digital reporting	L2	L2	L2	L2	**L4 R**	L4
12 Automation	L1	L2	L2	L1	L3	**L3 R**
13 Responsible AI	L2	L2	L2	L2	L3	**L5 R**
14 Human validation	L2	L3	L3	L3	**L4 R**	L4

Three observations from the profile shape.

Specialists are deep and narrow, and that is correct. A cost engineer at Level 4 in cost management and Level 2 in scheduling is properly configured. Pressuring specialists toward uniform Level 3 across fourteen competencies produces generalists who can sign nothing.

The manager role is defined by verification, not production. Note how many Level 4 accountabilities sit with the Controls Manager. Level 4 means signing others' work. This is why promoting the strongest producer into a manager role fails when the person has never been assessed at register 4 — producing and verifying are different competencies.

Responsible AI sits at Level 5 with the Head of Controls, and nowhere else. Someone must answer externally for how the function uses AI. Distributing that accountability across a team means nobody holds it.

> **Figure 4 — suggested diagram.** A radar chart with fourteen axes, overlaying three role profiles (Cost Engineer, Planning Engineer, Controls Manager) in distinct tones. The specialist profiles are visibly spiky; the manager profile is broader and flatter. Caption: "Depth is not a deficiency."

11. The Coverage Rule

Original framework.

Organisations assess individuals and then aggregate the scores into an average. The average is close to meaningless.

> **The Coverage Rule.** A controls team's operational capability is set by its **weakest covered competency**, not by its average level.

Why. Competencies are not substitutes. Eight excellent cost engineers cannot produce a quantitative schedule risk analysis between them. If nobody in the team holds Level 3 in project risk, contingency will be set by percentage — on every project, regardless of how strong the cost function is.

The average conceals exactly this. A team that is Level 4 in three competencies and Level 1 in three others averages the same as a team that is Level 2 or 3 across the board. The first team has three capability holes and does not know it; the second is uniformly adequate.

The second-order effect: competence investment follows competence. Organisations train where they are already strong, because that is where the internal trainers are, where the vocabulary is shared, and where the enthusiasm lives. The strong cluster gets stronger and the gap widens. This is a predictable dynamic, and naming it is usually enough to interrupt it.

> **Callout.** Ask a controls function where it invested in development last year. Then ask which competency it is weakest in. In most organisations these are not the same answer, and frequently they are opposites.

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12. The Coverage Index

Original metric.

The Coverage Rule needs a number to be actionable.

> **CI = (professionals at Level 3 or above in a competency) ÷ (concurrent projects requiring it)**

Level 3 is the threshold because Level 3 is the point at which someone can own the output on a project without supervision. Below that, the competency exists in the team but cannot be independently deployed.

Coverage Index	Reading	Operational consequence
≥ 1.0	Covered	Every project can be served independently
0.5 – 1.0	Shared	Covered by splitting people across projects; a single point of failure is forming
0.2 – 0.5	Thin	Most projects go without; the competency is applied reactively, after a problem
< 0.2	Absent	The competency exists on paper. It is not an operational capability

How to compute it honestly. Count only assessed Level 3 holdings, verified under the Evidence Standard. Self-declared levels inflate the index by roughly the amount that makes it useless. Individuals count in every competency where they hold Level 3, so total holdings will exceed headcount — that is expected and correct.

What the index is for. It converts a competency assessment into a resourcing and recruitment decision. A CI of 0.11 in project risk is not a training observation; it is a statement that eight of nine projects will set contingency without quantification this year.

> **Figure 5 — suggested diagram.** A horizontal bar chart, fourteen bars, one per competency, sorted descending by Coverage Index. A vertical reference line at CI = 1.0 and a shaded danger band below 0.5. The visual should make the long tail unmissable.

13. What AI changes

Four of the fourteen published competencies are AI competencies — predictive analytics, AI-enabled project controls, responsible AI and human validation. That is 29% of the competency set, which is a deliberate statement about where the discipline now sits.

What AI compresses. Register 2 — application. A machine can now produce many controls artefacts: a first-draft variance commentary, a coded cost ledger, a schedule-logic screen, an extracted contract summary. The production work that once occupied much of a junior controls role is genuinely compressible.

What AI does not touch. Register 3 — judgement. Choosing which EAC assumption the organisation is adopting, and defending it to a board, is not a task with a defined method. Nor is deciding that a model trained on the organisation's historical ledger has learned its historical optimism.

What AI raises the price of. Register 4 — stewardship. Someone has to set the autonomy levels, define the verification requirements, and answer for the function's AI use. That is new accountability sitting on the same senior layer that was already scarce.

Register	Effect of AI	Consequence for the framework
1 · Recall	Reduced value — the machine recalls better	Assess less of it
2 · Application	Substantially compressed	Assess the defence, not the artefact alone
3 · Judgement	Unchanged in value, **harder to develop**	Section 14
4 · Stewardship	Increased in value and scope	Develop deliberately; do not assume it arrives with seniority

> **Callout — the validation dependency, restated.** Competency 14 requires the competency being validated. An organisation deploying a predictive EAC model needs someone at Level 3 in forecasting to verify its output. Deploying the model because the organisation lacks forecasting competence is the precise inversion of the correct order.

14. The Apprenticeship Gap

Original framework. The most consequential competency question in the profession.

Judgement has never been taught directly. It has been a by-product.

The traditional pathway ran like this: a junior professional builds the artefact many times (register 2). Through repetition they encounter the cases where the method strains — the accrual that does not fit the rule, the progress claim that cannot be right, the schedule whose float is an artefact of a constraint. Those encounters accumulate into pattern recognition, which is what register 3 judgement actually is. Then they are trusted to choose the method (register 3), and eventually to set it for others (register 4).

AI removes much of the volume in the first step. The concern is not that juniors will have less to do. It is that **the volume of application work was the mechanism that produced judgement**, and removing the mechanism does not remove the need for the output.

An organisation that automates its junior controls work will have adequate output next year and no Level 3 professionals in eight.

This is a slow failure and it is already underway. Nothing breaks in the first years. The gap appears when the current Level 3 and Level 4 population retires and the succeeding cohort has never built a time-phased budget by hand.

Four responses. None is sufficient alone.

One — deliberate practice. Require manual production of core artefacts during development, even where AI could produce them faster. Accept the inefficiency as a training cost, and say so explicitly rather than pretending it is productive.

Two — verification as pedagogy. Make developing professionals verify AI output. This is the strongest response, because verification demands knowing what right looks like, and it maps onto real work rather than simulated work. Competency 14 becomes the training route rather than an additional burden.

Three — reverse the exercise. Supply an AI-produced artefact with a deliberate error and ask the professional to find it. This tests judgement directly and takes a fraction of the time of building from scratch.

Four — preserve the artefact trail. Keep the record of what the machine produced and what the human changed. That record is the richest development material an organisation will ever hold, and almost nobody is retaining it.

> **Callout.** The organisations that will hold senior controls capability in fifteen years are the ones treating junior development as a deliberate investment now, at a visible cost, rather than as something that used to happen by itself.

> **Figure 6 — suggested diagram.** Two parallel pathways from L1 to L4. The upper, historical path shows a wide band of repeated application work feeding into judgement. The lower, current path shows that band narrowed to a thread, with the judgement node faded and a question mark. Four labelled bridges reconnect the thread to the judgement node, one per response.

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15. Assessing a competency claim

The decision sequence. For a claimed level in a competency:

1. **Is there an artefact?** No → the claim fails. There is no assessment without an output.
2. **Did this person produce it?** Inherited or templated work establishes nothing. Ask what they changed and why.
3. **Can they defend it?** Apply the three questions from Section 4 — why this method, what is it resting on, what would change it.
4. **Does the defence hold under challenge?** Introduce a changed circumstance. A Level 2 answer restates the method; a Level 3 answer adapts it and says why.
5. **Has an independent Level 4 verified it?** Line-manager assessment alone does not satisfy the standard.
6. **Is the level consistent with the signing test?** What would you trust this person to sign unsupervised?

Evidence types, ranked by strength

Strength	Evidence	Note
Strongest	A live artefact from a real project, defended under challenge, independently verified	The standard
Strong	A live artefact plus a written basis, verified	Acceptable where challenge is impractical
Moderate	A simulated artefact from an assessment exercise, defended	Useful where the person has not yet held the accountability
Weak	Certification alone	Establishes registers 1–2; does not establish level
Weak	Line-manager attestation alone	Measures the relationship as much as the capability
Not evidence	Attendance, tenure, job title, self-assessment	Proxies, all of them

> **Callout — on self-assessment.** Self-assessment is a poor measurement and a good conversation. Use it to open a development discussion; never use it to populate a capability matrix. Its errors are not random, and they run in both directions: the strongest professionals routinely under-rate themselves.

> **Figure 7 — suggested diagram.** A decision tree following the six steps, with each "no" branch terminating in a labelled outcome — "no assessment possible", "assess the original author", "level capped at L2", "requires independent verification". Terminal nodes shaded by outcome type.

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16. Best practice — the ten rules of assessment

1. **Assess the competencies the role requires, not all fourteen.** Three or four, properly evidenced, beats fourteen guessed.
2. **Never assess without an artefact.** No exceptions, including for senior people.
3. **Assess the defence, not the output.** The machine can produce the output.
4. **Use two independent assessors on any Level 4 claim.** Level 4 means signing others' work; the bar should be visibly higher.
5. **Separate assessment from performance review.** Competence and performance are different, and merging them makes people defend rather than disclose.
6. **Record the characteristic failure, not just the level.** It is the most useful line in the record for the person's supervisor and for them.
7. **Publish the matrix to everyone being assessed.** A framework people cannot read is a rating system.
8. **Reassess when the method changes, not on a calendar.** Annual reassessment of a stable competency is administrative theatre.
9. **Compute coverage before designing training.** Development spend should follow the Coverage Index, not the enthusiasm of the strongest cluster.
10. **Keep the evidence file with the individual.** Competence belongs to the professional. Portability is the point.

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17. Implementation roadmap

Three phases, three gates, twelve months. Each gate is a demonstrated capability, not a completed document.

Phase 1 · Days 0–90 — Establish the standard and a baseline

- Adopt the Proficiency Ladder and the Evidence Standard without modification, so results are comparable across the organisation and over time.
- Define the required level per competency for each role, using Section 10 as the starting point.
- Assess one representative cohort — twelve to twenty people — using two independent assessors, and measure inter-assessor agreement.
- Do not assess everyone yet. The first cohort is calibration.

Gate 1. Two independent assessors reach the same level on at least 80% of claims. If they do not, the framework is not yet being applied consistently, and rolling it out further will produce numbers nobody trusts.

Phase 2 · Days 90–180 — Diagnose the organisation

- Complete assessment across the controls function, limited to role-required competencies.
- Compute the Coverage Index for all fourteen competencies against concurrent project load.
- Identify every competency below CI 0.5 and state its operational consequence in plain terms.
- Present coverage, not averages, to the executive.

Gate 2. A coverage profile exists, and every competency below CI 0.5 has a named consequence and an owner.

Phase 3 · Days 180–365 — Close the gaps deliberately

- Direct development spend by Coverage Index, lowest first.
- Establish verification-as-pedagogy: developing professionals verify AI output as their primary development route.
- Begin retaining the artefact trail — what the machine produced, what the human changed.
- Recruit only where development cannot close a gap in time. A CI below 0.2 usually cannot be developed out within a year.

Gate 3. No role-required competency sits below CI 0.5, and every developing professional has a named next artefact and a verifier.

> **Callout — the sequencing rule.** Calibrate before you measure, measure coverage before you design training, and direct training by coverage rather than by demand. Organisations that invert this train their strongest cluster further and discover the real gap during an incident.

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18. Worked case — the coverage gap

> **Illustrative composite.** Figures are arithmetically consistent and constructed to demonstrate the method. They are not drawn from a single named organisation.

Setting. An EPC contractor. Controls function of 24 professionals supporting 9 concurrent projects. The function was regarded internally as strong, and by most conventional measures it was: low turnover, experienced staff, a respected cost team.

The assessment. All 24 assessed under the Evidence Standard against role-required competencies, two independent assessors. Individuals hold Level 3 in more than one competency, so holdings exceed headcount.

#	Competency	At Level 3+	Coverage Index	Reading
2	Cost management	8	**0.89**	Shared
6	Performance measurement	6	**0.67**	Shared
8	Commercial and contract controls	6	**0.67**	Shared
1	Project controls governance	5	**0.56**	Shared
4	Earned value management	4	**0.44**	Thin
11	Digital reporting	4	**0.44**	Thin
5	Forecasting and EAC	3	**0.33**	Thin
14	Human validation	3	**0.33**	Thin
3	Planning and scheduling	2	**0.22**	Thin
10	AI-enabled project controls	2	**0.22**	Thin
12	Automation	2	**0.22**	Thin
7	Project risk	1	**0.11**	**Absent**
9	Predictive analytics	1	**0.11**	**Absent**
13	Responsible AI	1	**0.11**	**Absent**

What the average said. Mean coverage across the fourteen competencies was 3.4 professionals at Level 3 or above — a mean Coverage Index of 0.38. Presented as an average, this reads as a function that is stretched but functioning.

What coverage said. Ten of fourteen competencies sat below CI 0.5. Three were operationally absent. **With one professional at Level 3 in project risk across nine projects, eight projects were setting contingency without quantitative analysis.**

That is not a training finding. It is a statement that eight of the contractor's nine live projects carried a contingency figure derived from a percentage rather than from quantified risk exposure — and it connects directly to the Silent Failures set out in Publication 01 of this series.

How the gap formed. The function had invested in cost training for four consecutive years. Cost was already its strongest competency. The internal trainers were cost engineers, the shared vocabulary was cost, and the training proposals that reached approval were the ones with the most articulate advocates. Meanwhile project risk had one Level 3 professional and no advocate at all.

This is the second-order effect from Section 11 operating exactly as described: **competence investment follows competence.**

The remediation. Development spend was redirected by Coverage Index, lowest first. Over the following year:

Competency	CI before	CI after	Route
Project risk	0.11	**0.56**	Four cost engineers at L3 developed into risk quantification; one external hire at L4
Predictive analytics	0.11	**0.44**	Three developed via verification-as-pedagogy against an existing model
Responsible AI	0.11	**0.44**	Three developed; accountability consolidated at L5 with the Head of Controls

The lesson that generalises. The function was not weak. It was **uncovered**, in specific and identifiable places, and its own assessment method — an average — was structurally incapable of showing that. The diagnosis took four weeks. The organisation had been operating with the gap for years.

> **Figure 8 — suggested diagram.** A before-and-after Coverage Index bar chart, fourteen competencies, sorted by the "before" value. The three absent competencies rendered in an alert tone, with their "after" bars overlaid in the accent colour. A single annotation on project risk reading "8 of 9 projects, contingency unquantified".

19. Common mistakes and lessons learned

Mistake 1 · Assessing on adjectives. If two assessors cannot agree, the framework is measuring personality. Return to artefact, defence, verification.

Mistake 2 · Reporting the average. The average is the one statistic guaranteed to conceal a coverage gap. Report the distribution and the Coverage Index.

Mistake 3 · Pushing specialists toward uniformity. A cost engineer at Level 4 in cost and Level 2 in scheduling is correctly configured. Uniform Level 3 across fourteen competencies produces people who can sign nothing.

Mistake 4 · Promoting the best producer into a verification role. Producing is register 2 and 3; verifying others is register 4. They are different competencies, and the transition needs deliberate development rather than a title.

Mistake 5 · Treating responsible AI and human validation as policy. They are assessable capabilities with artefacts. Policy without capability is a document that describes what nobody can do.

Mistake 6 · Deploying a model to compensate for a missing competency. Validation requires the competency being validated. This is the single most consequential sequencing error available to a controls function.

Mistake 7 · Merging competency assessment with performance review. People defend rather than disclose, and the assessment becomes negotiation.

Mistake 8 · Assuming stewardship arrives with seniority. Register 4 is developed, not conferred. An organisation with no Level 5 in responsible AI has nobody who can answer for its AI use, regardless of how senior its people are.

The lesson beneath all eight. Every one is a measurement failure before it is a capability failure. Organisations do not usually lack competence in ways they understand; they lack it in ways their measurement method cannot show.

20. Future outlook — five shifts

One. Verification becomes the primary junior activity. As production compresses, the entry-level role shifts from building artefacts to checking them. This is a genuine change in what an entry-level controls job is, and job descriptions have not caught up.

Two. The evidence file becomes portable professional property. Professionals who hold a verified record of artefacts and defences will move between employers and jurisdictions on that record. Organisations that keep evidence files locked in internal systems will find they have made their own people less valuable and their own hiring harder.

Three. Stewardship becomes the scarce register. Registers 1 and 2 are increasingly served by tools. The constraint moves to the people who can set a standard and verify against it — a population that was already thin and is not being deliberately replenished.

Four. Responsible AI moves from policy to assessed competency. As auditors and clients begin asking who verified an AI-assisted output and on what basis, an unassessed responsible-AI capability becomes a commercial exposure rather than a governance preference.

Five. Coverage becomes a procurement question. Owners will increasingly ask contractors not how many controls staff they have, but which competencies they can cover on this project — and ask for the evidence.

> **Callout.** Each shift rewards the same investment: assessed, evidenced, portable competence with deliberate development of the senior register. None rewards headcount.

21. Key takeaways

1. A competency is a claim about what a person can produce and defend. No artefact, no competency.
2. The Evidence Standard is artefact, defence and independent verification. All three, every time.
3. An examination reaches recall, application and partly judgement. It cannot reach stewardship — and a certification body should say so.
4. Every proficiency level has a characteristic failure. Naming it is more useful than naming the strength.
5. Levels attach to competencies, not people. A spiky profile is correct.
6. Team capability is set by coverage, not by average. The average conceals exactly the gap you need to see.
7. The Coverage Index converts assessment into a resourcing decision: professionals at Level 3+ divided by concurrent projects.
8. Competence investment follows competence. Interrupt it deliberately, or the strongest cluster gets stronger while the gap widens.
9. Four of the fourteen competencies are AI competencies. Human validation requires the competency being validated.
10. Judgement was a by-product of volume, and AI removes the volume. Verification-as-pedagogy is the strongest available replacement.

22. The Competency Assurance Checklist

Twenty-eight tests. For a capability review, an internal audit scope, or contractor due diligence. Each is answerable **yes** or **no**; a qualified answer counts as no.

The standard 1. Is every competency claim supported by a named artefact? 2. Can the professional defend the method choice, the assumptions and what would change the answer? 3. Is every Level 3 and above claim independently verified? 4. Are Level 4 claims verified by two assessors? 5. Is the assessor at least one level above the claim being assessed? 6. Is the framework published to everyone being assessed? 7. Is assessment separated from performance review?

Consistency 8. Has inter-assessor agreement been measured? 9. Is agreement at or above 80%? 10. Are the same evidence types accepted across departments? 11. Is self-assessment excluded from the capability matrix? 12. Is tenure excluded as evidence? 13. Is certification treated as evidence of registers 1 and 2 only?

Coverage 14. Is the Coverage Index computed for every competency? 15. Is it computed against actual concurrent project load? 16. Does any role-required competency sit below CI 0.5? 17. Does any sit below CI 0.2? (A yes here is an operational absence, not a training gap.) 18. Is coverage — not average level — what the executive sees? 19. Has each competency below CI 0.5 got a named consequence? 20. Has each got a named owner?

Development 21. Does development spend follow the Coverage Index rather than internal demand? 22. Does every developing professional have a named next artefact? 23. Does each have a named verifier? 24. Is verification-as-pedagogy in deliberate use? 25. Is the artefact trail retained — what AI produced, what the human changed?

Stewardship and AI 26. Is there a named Level 5 for responsible AI? 27. Does everyone validating AI output hold Level 3 in the competency being validated? 28. Is register 4 developed deliberately rather than assumed to arrive with seniority?

> **Scoring guidance.** Fewer than 18 of 28: the organisation has a rating system rather than a competency framework. Items 1, 3, 14 and 27 are load-bearing — failing any one of them makes the remaining answers unreliable.

23. Key definitions and glossary

Term	Definition as used in this framework
Artefact	A tangible controls output that can be read and assessed — a schedule, a budget, a QSRA, a variance bridge
Assessment	Determination of a proficiency level against evidence, by a qualified independent assessor
Characteristic failure	The error mode that belongs to a proficiency level; what supervision should watch for
Competency	A claim about what a professional can produce and defend, assessable to a level
Coverage Index (CI)	Professionals at Level 3 or above in a competency, divided by concurrent projects requiring it
Defence	The professional's account of why this method, what assumptions it rests on, and what would change the answer
Evidence file	The individual's accumulated record of artefacts, defences and verifications
Human validation	Competent verification of machine-produced output, recorded with the reviewer named
Proficiency level	One of five positions — Assisted, Capable, Independent, Authoritative, Standard-setting
Register	One of four bands of competence — recall, application, judgement, stewardship
Role profile	The set of competencies and required levels for a defined role
Signing test	The placement heuristic: what can this professional be trusted to sign unsupervised?
Stewardship	Register 4 — setting method for others, verifying their work, accountability for the system
Verification	Independent confirmation of both artefact and defence by an assessor at Level 4 or above

Frameworks introduced in this publication

Framework	Section	Purpose
The Evidence Standard	4	Artefact, defence, verification — what makes a claim assessable
The Four Registers of Competence	5	What an examination can and cannot establish
The Proficiency Ladder	6	Five levels, each with its characteristic failure
The Competency Matrix	8	Fourteen competencies × five levels
Role Profiles	10	Required level by competency for six roles
The Coverage Rule	11	Capability is set by coverage, not average
The Coverage Index	12	The operational metric derived from the rule
The Apprenticeship Gap	14	How AI breaks the pathway that produced judgement
The Competency Assurance Checklist	22	Twenty-eight binary tests

24. References and further reading

Empirical research on project outcomes and capability - Flyvbjerg, B., Holm, M. S. and Buhl, S. (2002). "Underestimating Costs in Public Works Projects: Error or Lie?" Journal of the American Planning Association. - Flyvbjerg, B. and Gardner, D. (2023). How Big Things Get Done.

Competence, expertise and assessment - Dreyfus, S. E. and Dreyfus, H. L. (1980). A Five-Stage Model of the Mental Activities Involved in Directed Skill Acquisition. The novice-to-expert progression that underpins most professional proficiency scales. - Ericsson, K. A., Krampe, R. T. and Tesch-Römer, C. (1993). "The Role of Deliberate Practice in the Acquisition of Expert Performance." Psychological Review. The evidence base for deliberate practice referenced in Section 14. - Miller, G. E. (1990). "The Assessment of Clinical Skills/Competence/Performance." Academic Medicine. The "knows / knows how / shows how / does" pyramid — the closest established analogue to the four registers, drawn from medical education.

Professional standards and bodies of knowledge - AACE International — Total Cost Management Framework and Recommended Practices, including the cost estimate classification system. - Project Management Institute — A Guide to the Project Management Body of Knowledge (PMBOK Guide). - ISO 31000 — Risk management — Guidelines. - ISO/IEC 17024 — Conformity assessment — General requirements for bodies operating certification of persons. Referenced as the principle set against which PCI's certification framework is being developed.

AI governance - NIST — Artificial Intelligence Risk Management Framework (AI RMF 1.0). - ISO/IEC 42001 — Information technology — Artificial intelligence — Management system.

PCI AI publications - Knowing Early — An Executive Summary of the Project Controls Body of Knowledge. Knowledge Series 01. The Controls Value Chain, the Control Effectiveness Equation and the Seven Silent Failures referenced throughout this publication. - The PCL-AI Body of Knowledge — thirteen domains, sixty-one knowledge areas. - PCL-AI Examination Content Outline — the authoritative public syllabus.

25. About PCI AI

PCI AI · Project Controls Institute Global, Inc. is a certification body for project-controls professionals, incorporated as a Delaware Non-Stock Corporation.

The Institute certifies the **PCI AI Project Controls Leader (PCL-AI)**, one of three credentials in the PCI AI Project Leadership Certification Suite alongside the Project Finance Leader (PFL-AI) and the Project Management Leader (PML-AI).

This framework is published openly, including its explicit statement of what an examination cannot establish. A certification body that is candid about the limits of its own instrument is more useful to the profession than one that is not.

The governing principle applied throughout the Institute's work: **AI proposes; the professional disposes.**

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